VIDYAPEETH

BATCH CODE: 12-AN181MA

SUBJECT NAME: CHEMISTRY

CHAPTER NAME: Structure of atom

Lecture No. 03

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Today's Goal

Subtopic .. Isobars, Isotones, Isodiaphers, Isoesters

Subtopic ..

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Topic: Quick Last Lecture Revision - 01

* Z :→ Z = P

* A (mass number) = P + n

$n = A - Z$

Atom neutral

P = e
Z = P

Z = P = e → neutral atom

$^{16}_{8}O$

Z = 8
P = 8
e = 8

n = 16 − 8
= 8

* ions

$A \xrightarrow{-1e} A^{+}$

$A + 1e \longrightarrow A^{-}$

* N^{3-}

$^{14}_{7}N$

Z = 7
P = 7
e = 7 + 3 = 10
n = 14 − 7 = 7

Isotopes →

(a) physical properties = different

(b) chemical properties = same

Physical prop → A

chemical prop → Z

Isobars :→

Z = diff
A = same

→ atoms of different element having same mass number but different atomic number is called Isobars

eg: $^{14}_{6}C$ $^{14}_{7}N$ → Isobars

$^{40}_{18}Ar$ $^{40}_{20}Ca$ → Isobars

* Physical Prop = same
* chemical Prop = diff

Isotones →

→ having same no. of neutrons.

Ex.. Isotones

$^{23}_{11}Na$ $^{24}_{12}Mg$

neutrons = 12 neutrons = 12

Isotones

$^{19}_{9}F$ $^{20}_{10}Ne$

neutrons = 10 10

Isotones

$^{14}_{6}C$ $^{16}_{8}O$

n = 8 n = 8

Isodiaphers :→

→ having same number of (n-p) value.

Isodiaphers

20 Ne	40 Ca
10	20
p = 10	p = 20
n = 10	n = 20
(n-p) = 10-10 = 0	n-p = 20-20 = 0

Isodiaphers

14 N	16 O
7	8
n = 7	n = 8
p = 7	p = 8
n-p = 0	n-p = 0

Isodiaphers

7	19
3 Li	9 F
p = 3	p = 9
n = 4	n = 10
n-p = 1	n-p = 1

Isosters → In case of compound

- Same no. of atoms
- Same no. of electrons

Isosters

CO	CN^-
no. of atom = 2	= 2
no. of electrons = 6 + 8 = 14	(6 + 7) + 1 = 14.

• Isosters

CO_2	NO_2^+
no. of atom = 3	3
no. of $\bar{e}$ = 6 + 16 = 22	(7 + 16) − 1 = 22

* Atom → spherical

nucleus (p+n)
p → +
n → 0

→ orbit | shell | energy level

orbit → represented by mathematical number (1, 2, 3, 4 . . .)

shell → is represented by alphabets (K, L, M, N, . . .)

" Na : 2, 8, 1

orbit	shell
Ist	K
2nd	L
3rd	M
4th	N

$^{11}Na = 2, 8, 1$

Valence shell (outermost shell where $\bar{e}$ revolves)

Core shell (except valence shell all other shells)

$\boxed{\text{max}^m \text{ no. of } e \text{ in any shell} = 2n^2}$

n = no. of shell

	no. of $\bar{e}$
n = 1	$2 \times (1)^2 = 2\,\bar{e}$
n = 2	$2 \times (2)^2 = 8\,\bar{e}$
n = 3	$2 \times (3)^2 = 18\,e$
n = 4	$2 \times (4)^2 = 32\,e$

Subshell :→

→ Shell is divided into small parts called Subshell

* It is represented by s, p, d, f - - -

* **no. of subshell in any shell = n**

shell

Shell	subshell		
n = 1	1	s	1s
n = 2	2	s, p	2s, 2p
n = 3	3	s, p, d	3s 3p 3d
n = 4	4	s, p, d, f	4s 4p 4d 4f

★★★★★

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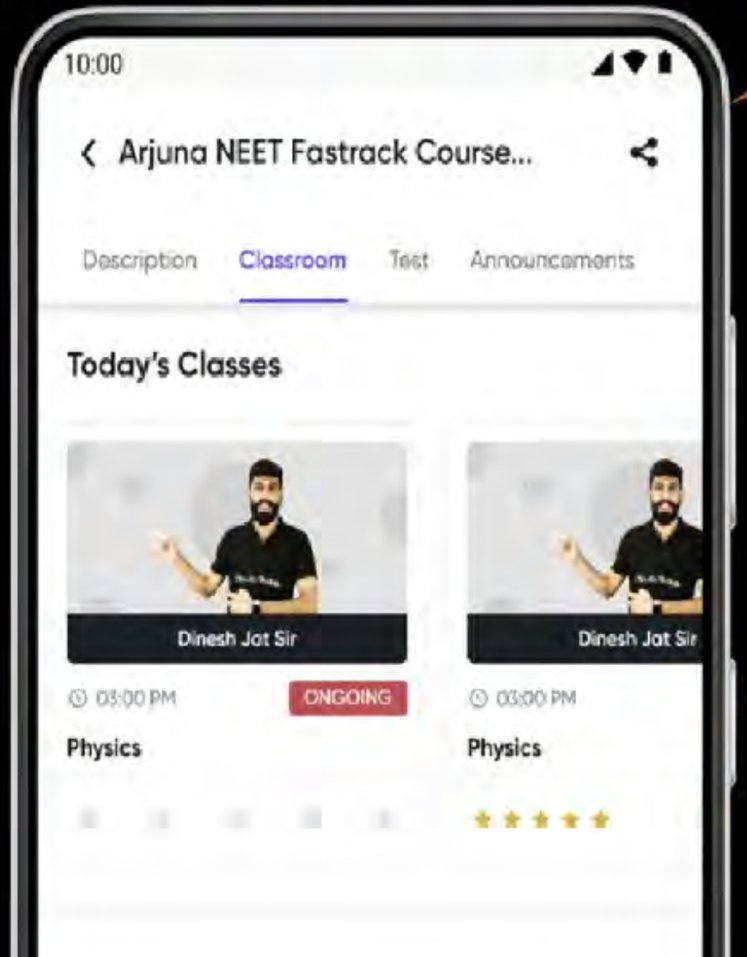
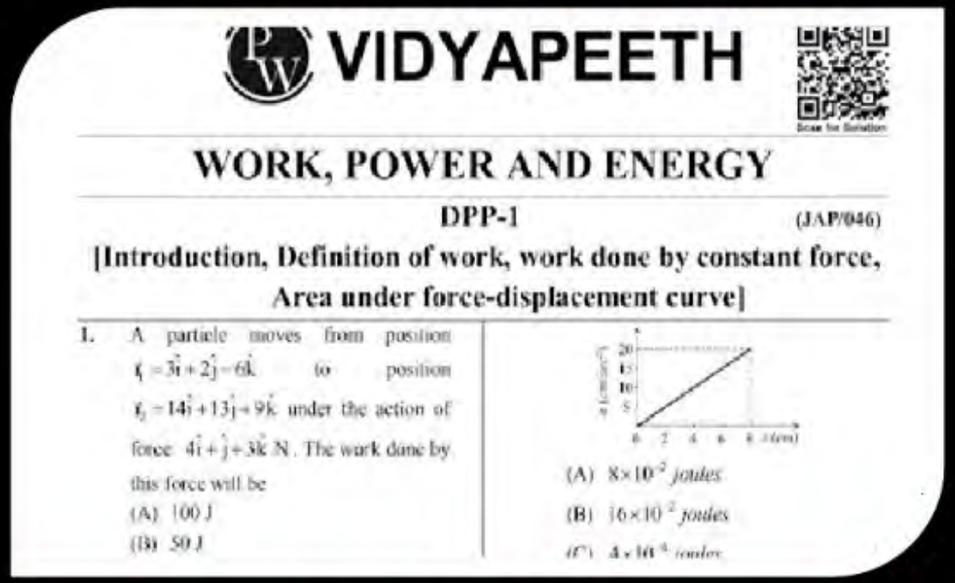


"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



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WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} joules

(B) 16×10^{-2} joules

Thank You!!!!

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